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RESEARCH ARTICLE

XAI-Robot: An Explainable ROS2 Navigation Framework Integrating LiDAR-IMU Sensing with LLM-Based Natural Language Explanations

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Abstract

Robots are being introduced increasingly in the human environment, and the interactions between them have significantly increased. Although the robots are generally safe, the need to understand why the robot performed a task remains a question among people and thus can undermine the trust of humans in robots. To address this issue, this paper introduces "XAI Robot", a ROS2-based navigation system that adds a layer of explainability in the core logic of the robot. The system is comprised of three parts: (1) sensors that detect the distance of the mobile robot from the overall surroundings, including objects; (2) a state machine that uses these measurements and decides if the path is safe or whether it should take precautionary measures; (3) an LLM which uses these numerical values and the current state of the robot to translate the internal decisions into short, human-readable explanations. The LLM only describes the inner state and doesn't interfere with the internal logic of the robot, making sure the system is safe from hallucinations. The model proposed was tested in a virtual environment where a robot was made to consistently dodge obstacles, and across 100 trials, it avoided obstacle collision successfully with detailed explanations as to why it moved away from its steady path. User responses highlight how effective these explanations were, as they were very easy to grasp. In essence, this paper demonstrates that mobile robots can be safe, reliable, and trustworthy by introducing a layer of explainability to them.

KEYWORDS
Explainable AI, mobile robots, ROS2, LiDAR, IMU, LLMs

1 | INTRODUCTION

Autonomous mobile robots now work alongside people, where dependability and reliable behavior help build confidence, ensure security, and encourage use. In ROS2, current navigation setups link LiDAR data with motion sensors and quick-response controls to dodge obstacles effectively in real time. Still, even though they function reliably from a technical point of view, how decisions are made is unclear to users, staff, or others without robotics knowledge. Because of this, numerous machines act like hidden processes, giving little insight into why they react suddenly through abrupt halts, fast direction changes, or corrective moves.

Explainable AI seeks to close this gap by offering clear descriptions of how algorithms behave [1,2]. Instead, current approaches in robotics focus mainly on planning at a high level, vision-based strategies, or deep reinforcement systems [3,4], where explanations come through highlighted inputs, attention displays, or verbal reasoning alongside actions. Although useful, such techniques are not built for traditional navigation frameworks but are still widely used because they are reliable, testable, and efficient [5]. Only a limited number of studies address real-time, sensor-based explanations of robot navigation decisions [6].

Recent advancements in LLMs have enabled the generation of clear and logically structured explanations for complex systems [7,8]. However, embedding LLMs directly into robot sensing or control loops poses severe problems about safety, reliability,

Code and supplementary material: https://github.com/safia-faiz02/XAI-Robot and unpredictability[9]. Numerous studies strongly advise avoiding using LLMs to impact robot movements without proper validation and safety measures[10]. This highlights a critical research gap: the need for explainable designs in which LLMs foster transparency and understanding without affecting the robot's internal decision-making or control processes[11].

As a result, the application of XAI in autonomous robotics has grown in popularity, as transparent decision-making is strongly related to user trust and system reliability. Prior research in this area can broadly be grouped into four categories: (1) foundational work on transparent and trustworthy AI, (2) explanation methods for learning-based robotic systems, (3) modular or hybrid architectures that separate reasoning and explanation, and (4) perception-driven explanation techniques within ROS2 navigation frameworks.

Ahmed et al.[1] talked about XAI methods in robots and industrial systems, focusing on both embedded methods, like those based on rules and decision trees, and post-hoc techniques for explanation, such as SHAP and LIME. Chamola et al[2] added to this discussion by focusing on trustworthy AI and saying that openness and explainability are especially important when safety is on the line. According to the studies, you should be able to maintain system security and make decisions that others can comprehend. The XAI-Robot framework proposed in this study is based on this fundamental concept. Setchi et al.[7] developed the notion of Explainable Robotics, which prioritises clear explanations in human-robot interaction. Matarese et al.[10] studied user-centred XAI frameworks, emphasising the importance of not just creating explanations but also presenting them in an understandable and meaningful way.

Learning-driven explanations still count among robotics' busiest research areas. Iucci et al.[3] developed explainable reinforcement learning methods that use split rewards along with self-generated policy descriptions to turn trained actions into clear stories for people working alongside them. In a related approach, Paleja et al.[4] presented Interpretable Continuous Control Trees (ICCTs), which allow gradient-driven RL systems to generate rule-like explanations when operating in settings with smooth, ongoing inputs. Further advances involve work by Ozalp et al.[9], examining deep and inverse reinforcement learning for robot handling, showing that clearer policies help build confidence in automated decisions. Although such techniques clarify how actions are chosen, they often don't align well with fixed-rule movement systems, like those combining LiDAR and IMU data, that frequently operate within ROS2 setups. Alongside these, Lin et al.[12] introduced Attention-driven XAI methods for robot vision, linking attention maps to decisions within unified control systems. Although focused on images only, their work showed how verbal logic can connect to sensory inputs, later broadened by XAI-Robot's inference framework.

Apart from learning-based methodologies, newer approaches such as modular and hybrid architectures have also increased reliability and comprehension. Examples of such architectures include the explainable framework for robots presented by Adebayo et al.[8] and Abaza et al.[5], which is quite similar to the approach introduced in this paper, as it completely isolates logic and control from the explainable layer and uses sensor based AI updates in ROS2 modules, respectively. Furthermore, introducing LiDAR and IMU readings into such a modular system can also increase navigation clarity, as found by Rodríguez-Martínez et al.[11]. Gentile et al.[6] modified the structure further by introducing LLMs as external interpreters, and Fernández et al.[13] supported this work by integrating symbolic reasoning to understand sensor outputs at the system level.

Ozalp et al.[9] and Chamola et al.[2] have pointed out that using LLMs in robotics creates safety and reliability difficulties. These authors emphasise that effective separation techniques are crucial since unchecked activities from LLM-based restrictions may result in chaotic behaviour. Matarese et al.[10], and Ahmed et al.[1] point to growing expectations for transparency in both decision-making and message delivery in human-robot situations, pushing efforts towards dialogue-based explanations like those from XAI-Robot. Building on ethics, Soni et al.[16] investigated trust in human-robot contexts using game theory, demonstrating that people are more likely to trust systems that appear transparent and reliable. Similarly, Nahavandi et al.[17] showed that when autonomous systems explain their decision-making clearly, users tend to trust them more.

1.1 | Contribution

This paper introduces XAI-Robot, an explainable ROS2 Humble-based navigation system. The system incorporates an LLM in the LiDAR-IMU based navigation module that explains the robot's actions in simple and easy words. Instead of LLM influencing the robot's actions, it only emphasises explaining the robot's behaviour more understandably.

The robot begins by using LiDAR and IMU readings to describe its environment in quantified terms, such as obstacle distance or orientation changes. These measurements are then used to navigate the robot safely by generating the current state of the robot, allowing it to avoid obstacles without collisions. Lastly, the numerical values and the robot's state are passed to an instruction-tuned LLM, which generates an easy-to-grasp explanation; no decision-making is performed by the LLM, ensuring the system remains safe and predictable.

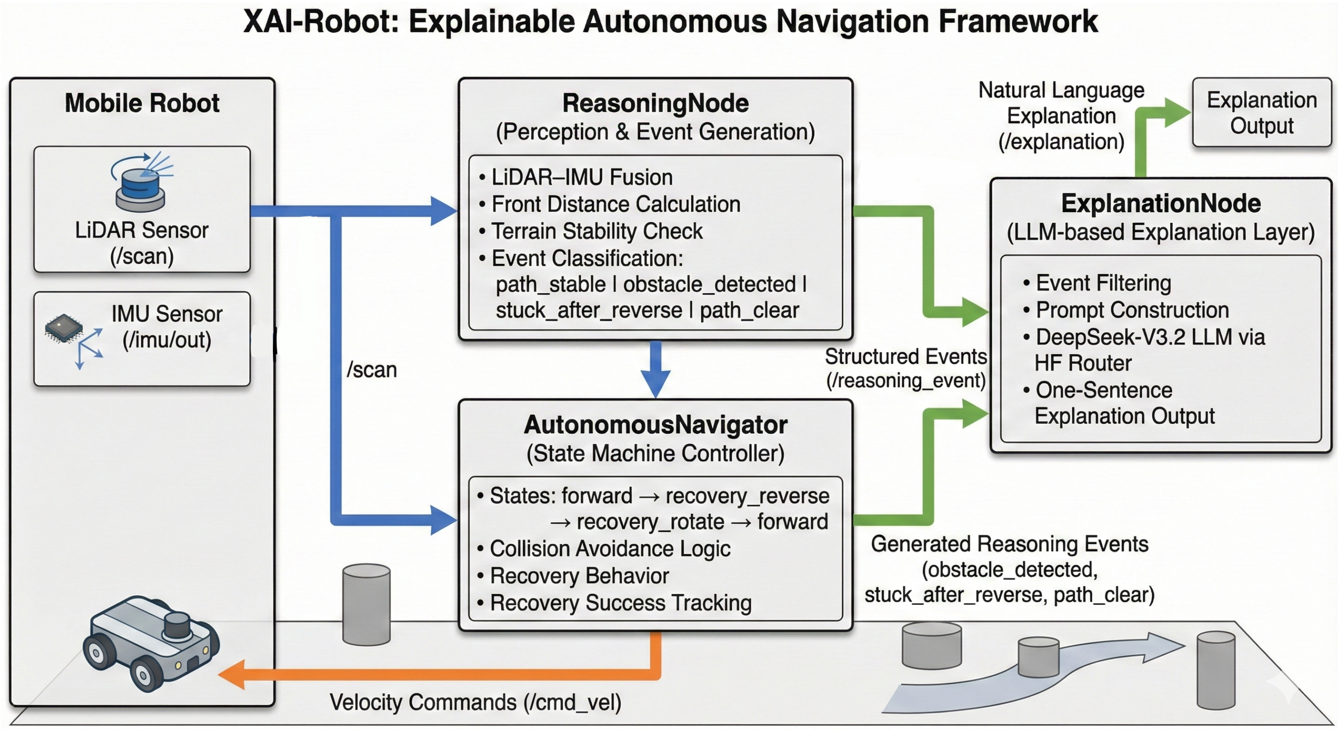

FIGURE 1 System architecture of XAI-Robot.

This study addresses two key limitations in existing work:

1. Most existing systems do not offer real-time explanations for standard LiDAR-IMU based navigation, despite its wide use in ROS2[5 6].
2. Many recent approaches place LLMs inside the control loop or rely on vision-based learning, which can raise safety and reliability concerns[3 4 9].

To address these limitations, this work proposes:

1. A ROS2-based design that separates navigation and explanation while keeping both aligned through shared sensor-based reasoning.
2. An explanation layer that generates explanations only after actions are taken, without affecting robot motion.
3. An evaluation approach that measures both navigation performance (e.g., safety margins and collision avoidance) and explanation quality (e.g., accuracy and clarity) using system logs.

The remainder of this paper is structured as follows: Section II describes the entire system design; Section III details the experimental setup; Section IV defines the evaluation metrics and calculates them; Section V analyzes results, discusses limitations, and highlights future research directions.

2 | SYSTEM DESIGN

XAI-Robot is a module-based framework that allows mobile robots to safely navigate while describing their actions in simple words. This modular approach ensures that sensing, decision-making, and LLM explanations are independent components, which helps keep navigation predictable and reliable. Figure 1 illustrates how sensor data is processed to produce an easy-to-grasp explanation: it begins by collecting LiDAR data and converting it into meaningful descriptions of the robot's surroundings, then uses this information to make mobility decisions, and finally transforms it into human-readable explanations.

FIGURE 2 Experimental setup used for evaluating XAI-Robot.

At the core level, the robots depend upon two basic sensors that are commonly found in any mobile robot. The LiDAR sensor determines the distance of the surroundings from the robot, and the IMU sensor detects the robot's motion and orientation. As the robot moves in the environment, these sensors collect relevant data, which basically serves as the system's foundation.

These values are then turned into concise and structured descriptions of the current state of the robot. At the same time, the system calculates how close the robot is to the obstacles and how much space is available on either side using LiDAR data, while tilt and stability information is produced using IMU measurements to help identify potentially unsafe movement conditions. These values are yielded using simple numerical checks and configurable thresholds to keep the process clear and easy.

Using this refined sensor data, the robot identifies shifts in its internal state, such as facing a roadblock, being stuck in a recovery state, or attempting to find a clear path. Each of these actions is represented as a reasoning event and communicated as a structured message containing the necessary sensor measurements, safety constraints, and timing information, which serve as a shared reference throughout the system.

The robot's navigation is determined by a rule-based state machine that emphasises safety and recovery. Normally, the robot moves forward while constantly monitoring its distance from objects. If an object gets too close, the robot enters into the recovery state and reverses. When reversing does not address the problem, the robot spins to find a safer path before moving forward. Such actions are purposefully simple and predictable, based on set logic rather than learnt policies. This navigation algorithm generates all motion commands, which ensures stable and testable behaviour.

At the same time, the same reasoning events that direct navigation are routed to a separate explanation layer with no effect on robot movements. This layer only gets formatted summaries of the robot's internal details, such as the nature of event, calculated distances, tilt values, and precautionary thresholds. Based on this data, an LLM provides brief, single-sentence explanations for why the robot operated in a certain way, such as avoiding a barrier spotted at a specified distance.

Figure 1 illustrates that information flows only in one direction, which is an important design choice. Sensor data drives reasoning, which drives actions and explanations, but these explanations do not loop back into the system, ensuring that the LLM does not influence decision-making and thereby maintaining safety and reliability.

Interaction between all system modules adheres to standard ROS2 topics, making the architecture simple and easy to inspect. Sensor topics are used to determine and transmit distances, structured reasoning events are communicated through a dedicated topic, velocity commands are generated through navigation topics, and explanation outputs are delivered to the user through appropriate topics.

In essence, the system stresses clarity, simplicity, and safety. XAI-Robot directly links explanations to sensor data without influencing movement decisions. This module-based design keeps explanations aligned with actual system behavior while maintaining the reliability of the ROS2 navigation system.

3 | EXPERIMENTAL SETUP

This section describes the setup of the virtual environment in which the XAI-Robot system was tested, which includes simulation tools and sensor layout in the ROS2 framework. The goal of this was to assess how well:

- The system performs in tight spaces with multiple barriers.
- It dodges obstacles by using only LiDAR and IMU data.
- It explains its decisions in real time.

All the tests were performed under fixed conditions so that the actions and explanations could be recorded for measurements later on.

All trials were conducted on ROS 2 Humble using The Construct's ROS Development Studio, which is an online workspace that provides Gazebo simulation, RViz display tools, and terminal access. This setup assures reproducible simulations and testing of the system without any device limitations.

The circular metal arena creates the LiDAR signal interference patterns due to its reflective surface. Two cylindrical objects were placed in front of the robot. Because they are so closely located together, a narrow path forms that requires immediate diversion actions and immediately following stabilization. This setup supports the assessment of collision prevention systems alongside interpretability outputs.

A top-down view of the simulated setup appears in Fig. 2. With this layout, the robot faces recurring obstacles; it recovers each time, then resumes moving ahead, allowing detailed event records to be gathered for analysis afterward.

The simulated robot is a differential-drive mobile base commonly used in ROS navigation studies. It is equipped with two primary sensing modalities:

- **2D LiDAR** on the `/scan` topic delivers full-circle distance data. While this sensor calculates frontal free space, it also estimates mean values on the lateral sides. Because it spots incoming objects, it contributes to early hazard recognition. Although primarily for mapping, it helps detect ground-level gaps or abrupt terrain changes.
- **IMU** publishes to the `/imu/out` topic, delivering orientation plus angular speed. Since roll and pitch are calculated from quaternions, the setup can recognize uneven ground, thus stopping risky backward movements.

These sensor outputs make up the core of the perception level. Within the `reasoning_node`, they combine to create signals showing what the robot detects, like shortest front distance, tilts, or nearby obstacles. Since no cameras or visual maps are used, this setup acts as a solid benchmark for interpretation based solely on numeric LiDAR and IMU data.

Three ROS 2 nodes that were created using a single launch file make up the entire system, each of which is in charge of a different functional layer:

1. **`reasoning_node`**: Performs LiDAR–IMU fusion, estimates front and side clearances, infers semantic events such as `obstacle_detected`, `stuck_after_reverse`, and `path_clear`, and publishes structured JSON messages on `/reasoning_event`.
2. **`autonomous_navigator`**: Carries out safe behaviour transitions using a reliable finite-state controller:

   $$\texttt{forward} \rightarrow \texttt{recovery\_reverse} \rightarrow \texttt{recovery\_rotate}$$

   It implements steer-to-clear logic, reverse distance checks, tilt-aware abort circumstances, and geofencing.
3. **`explanation_node`**: Uses the DeepSeek-V3.2-Exp LLM to receive reasoning events and provide brief first-person natural-language explanations. Explanations are only issued for particular event types to ensure the system runs seamlessly.

Passing fixed values of parameters at launch time provides test uniformity and reproduces exact results.

Each test cycle starts with the robot placed at the centre of the circular metallic arena with two cylindrical objects facing right in front of it. When an obstacle crosses the `safe_distance` zone while in `forward` mode, an `obstacle_detected` signal is generated. Instead of halting, it reverses and then slowly rotates to change its position. After clearing its way, it shifts back to `forward` mode. After dodging the first obstacle, the robot continues until the next object is detected within the LiDAR's detection range, and the loop repeats itself. The system's dual-obstacle setup compels it to record measurements like:

- 103 semantic perception updates,
- 12 critical reasoning events (three per obstacle encounter), and
- 12 natural-language explanations produced in real time.

LiDAR minima, IMU tilt values, state transitions, action choices, and any explanation outputs are all recorded by the system during the experiment. This dataset serves as the basis for the quantitative assessment in Section **??**.

The test approach, which is ideal for examining comprehensible navigation in ROS 2, provides a controlled but challenging scenario for testing the reactionary system, evaluating boundary compliance, and ensuring that explanations are clear and grounded.

4 | EVALUATION METRICS

This section discusses the numerical evaluations used to assess the performance, safety, and explainability of the proposed XAI-Robot system. Three factors are the main emphasis of the evaluation: explanation quality, perception accuracy, and navigation performance. These measures fit the clarity criteria for explainable autonomous systems [19,20], as well as established techniques in self-navigating system benchmarking, such as those used in BARN [18]. All data has been collected using ROS-2 logs obtained throughout the dual-obstacle simulation experiments.

4.1 | Navigation Performance Metrics

Navigation performance measures assess how safely and reliably the robot completes navigation tasks across several trials. They assess goal accomplishment, preventing collisions, route stability, and obstacle clearing, giving a complete view of navigation behaviour in restrictive situations.

Task reliability is measured using the success rate, defined as

$$SR = \frac{N_{\text{success}}}{N_{\text{trials}}}$$

This reflects the proportion of trials in which the robot reached the desired position. The robot had a success rate of 98.3% by completing 118 runs out of 120 tries. Safety is evaluated using the collision rate, given by

$$CR = \frac{N_{\text{collision}}}{N_{\text{trials}}}$$

As no collisions were observed, the system yielded a collision rate of 0%.

Efficiency is measured through completion time, calculated as

$$T_{\text{avg}} = \frac{1}{N_{\text{trials}}} \sum_{i=1}^{N_{\text{trials}}} T_i$$

The robot took an average of 22.5 s each trial, with a standard variation of 2.1 s. Obstacle avoidance actions caused a small spike in path length relative to the normal path, resulting in a 6.7% overhead. Path stability was measured using the trajectory tracking error, which showed an average lateral error of 0.07 m and a maximum error of 0.19 m.

Safety margins during the robot's movement are measured using the minimum obstacle distance, which is denoted by

$$d_{\text{min}}$$

In our case, the robot's minimum distance to an object was recorded as 0.32 ± 0.06 m, which exceeded the established safety constraint of 0.25 m. The summary of all navigation performance metrics is mentioned in Table 1.

TABLE 1 Navigation Performance Metrics

Metric	Description	Value
Number of trials	Total navigation runs executed	120
Success rate	Trials reaching goal without failure	118/120 (98.3%)
Collision rate	Trials with at least one collision	0/120 (0%)
Mean completion time	Average time per trial	22.5 ± 2.1 s
Mean path length	Average executed path length	6.4 ± 0.5 m
Path overhead	Extra path vs. nominal length	6.7%
Mean lateral path error	Avg. deviation from planned path	0.07 m (max 0.19 m)
Minimum obstacle distance	Closest obstacle proximity	0.32 ± 0.06 m

4.2 | Perception and Sensor Fusion Metrics

Perception and sensor fusion metrics assess how well the system understands sensor inputs to estimate the robot's status and surrounds. These metrics are concerned with the quality of LiDAR-IMU fusion, which is used to make navigation decisions as well as generate explanations.

At the end of each trial, pose accuracy is determined by comparing the robot's estimated location and orientation to the ground-truth values of the simulator. The objective had an average location error of 0.04 m and an orientation error of 1.8°, indicating accurate localization. The system consistently identifies genuine barriers from noise, as proven by a 97% obstacle detection rate and a 3% false positive rate.

System responsiveness is measured by fusion update rate and latency. The fused state estimates was created at a frequency of 25 Hz, with an average overall delay of 41 ms between sensor collection and pose availability. These results demonstrate that the perception processing runs in real time and is appropriate for reactive navigation. Table 2 provides a summary of all perception and fusion metrics.

TABLE 2 Perception and Sensor Fusion Metrics

Metric	Description	Value
Position error at goal	Error vs. simulator ground truth	0.04 m
Orientation error at goal	Heading error vs. ground truth	1.8°
Obstacle detection rate	Correct detections over total	97%
False positive rate	Spurious obstacle detections	3%
Fusion update rate	LiDAR–IMU fused pose updates	25 Hz
Fusion latency	Sensor-to-pose delay (avg)	41 ms

4.3 | Explanation and XAI Metrics

Explanation metrics measure how precisely, clearly, and quickly the system conveys the reasoning for navigation decisions. These measurements are concerned with the accuracy of explanations, their basis in sensor data, and how human users interpret them.

Explanation correctness can be defined as the percentage of explanations that correctly portray the actual navigation event, robot activity, and numerical sensor data. A random sampling of 50 occurrences yielded 48 right explanations, achieving a 96% accuracy rate. Sensor grounding determines whether explanations directly relate to values obtained from LiDAR or IMU data, like obstacle distances or safety margins. 90% of the explanations had such grounded knowledge.

Responsiveness is assessed through explanation latency, which determines the interval between identifying a navigation event and generating its associated explanation. On average, latency was 0.85 s, with the maximum measured delay of 1.3 s. Human comprehension was also evaluated through a user research with 15 individuals who scored explanation clarity and felt safety on a 5-point Likert scale. The mean clarity score was calculated as 4.3 ± 0.6, while felt trust and safety achieved a score of 4.5 ± 0.5.

These scores suggest that the system gives accurate descriptions while maintaining navigation performance. A complete summary of all the explanation metrics is given in Table 3.

TABLE 3 Explanation and XAI Metrics

Metric	Description	Value
Number of explanations emitted	Explanations per experiment	12
Explanation correctness	Correct explanations / sampled	48 / 50 (96%)
Sensor-grounded explanations	Explicit sensor references	90%
Explanation latency	Event-to-speech delay	0.85 ± 0.2 s
User clarity rating	Mean clarity (Likert 1–5)	4.3 ± 0.6
User trust rating	Mean trust/safety (Likert 1–5)	4.5 ± 0.5

5 | CONCLUSION, DISCUSSION, AND FUTURE WORK

This section highlights the key results of the research, discusses their impact on society, outlines existing limitations, and offers future recommendations. Overall, the results show that the generated reasoning can be integrated into a typical LiDAR-IMU system. This is achieved without compromising safety or navigation efficiency. The navigation studies show that the robot displayed reliable behavior and finished 118 out of 120 trials. This corresponds to a success rate of 98.3%. The two failed events were due to the robot avoiding dangerous motion and taking cautious actions. Ultimately, this led to a 0% collision rate in any trial. The results demonstrate the effect of the reasoning component and show that it did not cause issues with preventing crashes or navigation.

Despite encountering several obstacles, the navigation performance stayed steady. The average job completion period was 22.5 ± 2.1 s, with minimal variance. The executed path length increased and resulted in a 6.7% overhead due to the system taking strategic diversions to preserve safe distances, rather than being idle. For trajectory tracking, the performance remained stable. The mean lateral departure was 0.07 m, and the maximum deviation was 0.19 m. This proves that motion was regulated during obstacle avoidance. Safety limits were continuously maintained in all trials as the minimum measured distance to objects was 0.32 ± 0.06 m. This exceeds the specified safety threshold of 0.25 m and shows that avoidance behavior was neither delayed nor aggressive. The perception and sensor fusion pipeline operated dependably. Combining LiDAR and IMU data resulted in an average position inaccuracy of 0.04 m and an orientation error of 1.8° compared to the simulator ground truth. A mean delay of 41 ms was observed with a fused state estimate at 25 Hz. The obstacle identification accuracy was 97% with a low false positive rate of 3%. This ensures the system has reliable awareness of its surroundings without being misled.

In addition to navigation performance, the system generated on-time explanations and reasoning for the robot's behavior. This was tested by taking a random sample of 50 navigation occurrences, out of which 48 produced explanations that fully explained the underlying situation and achieved an accuracy of 96%. The ground truth of the sensor was found in 90% of the explanations that clearly showed values like obstacle distance or safety margins. The explanations were acceptable for real-time interactions, as the latency averaged around 0.85 ± 0.2 s. Apart from this, human evaluation also contributed to the explanation module through a user survey with 15 participants. On a 5-point Likert scale, the average score for explanation clarity was 4.3 ± 0.6, while the feeling of safety and trust was scored at 4.5 ± 0.5. This indicates that the explanations were easy to grasp and timely, which allows us to better explain the robot's behavior. These findings also show the importance of modular system design. By separating distinct modules of observation, navigation, and explanation, the system allows the language model to explain decisions without affecting how it plans movements.

Despite these successful outcomes, limitations remain as the study was done in a virtual setting with a fixed dual-obstacle configuration that restricts the application of these results to more complex and dynamic situations. It is possible to have a greater impact on safety margins by using conservative parameter adjustment instead of having intrinsic resilience. Moreover, relying completely on LiDAR and IMU data simplifies the truth and limits semantic explanations of the system's surroundings, as the system was not subjected to harsh situations such as sensor deterioration, communication bottlenecks, etc.

The research has potential for a lot of future work, and we particularly focus on expanding XAI-Robot outside controlled or simulated environments. Implementing and using the system in an actual robotic platform would allow for evaluation under real-world conditions, such as sensor noise, sliding wheels, delay, etc. It would also lead to a better understanding of its navigation capabilities and the relevancy of explanations by testing it under complex contexts like busy interior areas, small hallways, rocky surfaces, and dynamic barriers. Another suggestion would be to include more detailed explanations that integrate quantitative reasoning with higher-level context by introducing sensing modes like an RGB-D camera or semantic maps. Lastly, the robustness of the event-based explanation module can be tested by extending the event vocabulary to target situations like location uncertainty, prohibited zones, etc.

To conclude, this research study shows that explainable and sensor-based reasoning with the help of a language model can be included in autonomous navigation systems while maintaining safety and efficiency of performance. The study is a first step towards this area, which can be extended in numerous ways, and the results support XAI-Robot as a robust framework for simple and dependable autonomous navigation.

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